

Dharmsinh Desai University
Data Structures
Graph Implementation and Traversals

Exercise:

- Create Python programs for the following.
- Create your programs in the VS Code with complete documentation.

1. Write a program to implement an undirected graph with the following.
 - a. Create an adjacency matrix.
 - b. Print the information of the graph such as number of edges, edges list, degree of each vertex.
 - c. Implement traversal of graphs using DFS.
2. Write a program to implement an undirected graph with the following.
 - a. Create an adjacency List.
 - b. Print the information of the graph such as number of edges, edges list, degree of each vertex.
 - c. Implement traversal of graphs using BFS.

Note: Include output for 2 graphs (having at least 5 vertices) in each program. Also show a visual graph in your written submission.

Practice Exercise:

1. Write a program to implement a directed graph with the following.
 - a. Create an adjacency matrix.
 - b. Print the information of the graph such as number of edges, edges list, degree of each vertex.
 - c. Implement traversal of graphs using BFS.
2. Write a program to implement a directed graph with the following.
 - a. Create an adjacency List.
 - b. Print the information of the graph such as number of edges, edges list, degree of each vertex.
 - c. Implement traversal of graphs using DFS.